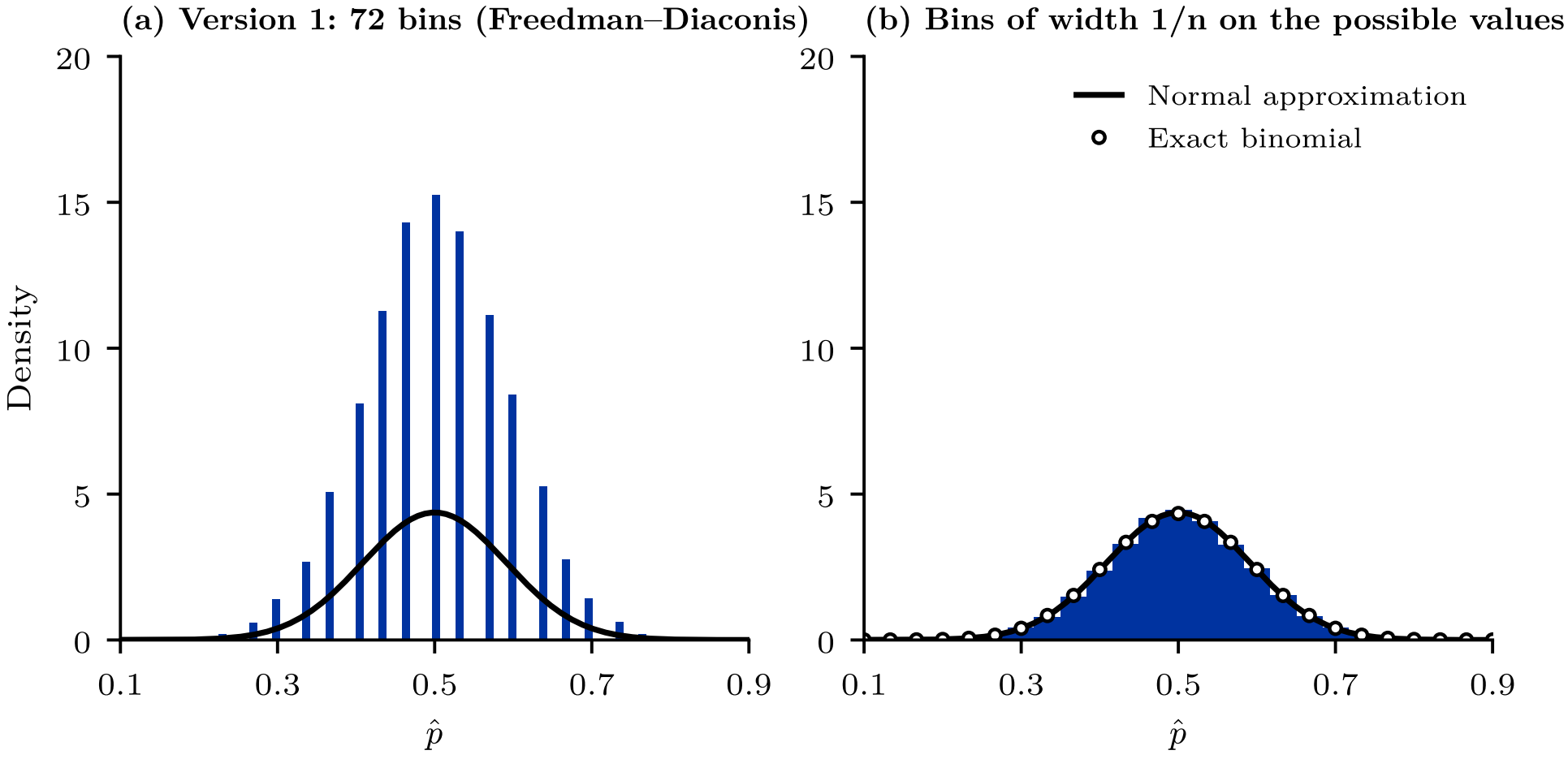
(a) Version 1: 72 bins (Freedman–Diaconis)
(b) Bins of width 1/n on the possible values
Density
20
15
10
5
0
0.1
0.3
0.5
0.7
0.9
$\hat{p}$
Normal approximation
Exact binomial